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Composite compaction system supported by a number of morphable bag supports

Abstract

Composite compaction system and method of use are presented. A method of supporting a composite compaction system is presented. A number of morphable bag supports of the composite compaction system is held in a retracted state. A contact face of a vacuum bag of the composite compaction system is placed in contact with a composite material on a support structure. The number of morphable bag supports is placed in an engaged state to conform to the support structure.

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Background/Summary

BACKGROUND INFORMATION

- Field
(1) The present disclosure relates generally to composite processing and more specifically to methods and apparatus for performing compaction on a composite material.
- Background
(2) Composite materials are strong, light-weight materials created by combining two or more functional components. For example, a composite material may include reinforcing fibers bound in polymer resin matrix. The fibers can take the form of a unidirectional tape, woven cloth or fabric, or a braid.
(3) In manufacturing composite structures, layers of composite material are typically laid up on a tool. The layers may be comprised of fibers in sheets. In some cases, resin may be infused or pre-impregnated into the sheets. These types of sheets are commonly referred to as prepreg.
(4) After laying up a layer of composite material on a tool, the composite material can be compacted onto the tool. Compaction can flatten and remove air and wrinkles from the composite material. It can be undesirably difficult to apply conventional composite compaction films over objects, especially those with a contour.
(5) Therefore, it would be desirable to have a method and apparatus that takes into account at least

some of the issues discussed above, as well as other possible issues.

SUMMARY

(6) An embodiment of the present disclosure provides a method of supporting a composite compaction system. A number of morphable bag supports of the composite compaction system is held in a retracted state. A contact face of a vacuum bag of the composite compaction system is placed in contact with a composite material on a support structure. The number of morphable bag supports is placed in an engaged state to conform to the support structure.

(7) An embodiment of the present disclosure provides a composite compaction system. The composite compaction system comprises a vacuum bag having a contact face configured to contact a composite material and a support face opposite the contact face, and a number of morphable bag supports formed of a flexible polymeric material connected to the support face and configured to morph between a retracted state and an engaged state.

(8) An embodiment of the present disclosure provides a composite compaction system. The composite compaction system comprises a vacuum bag having a contact face configured to contact a composite material and a support face opposite the contact face, and a number of morphable bag supports formed of a flexible polymeric material connected to the support face and configured to morph between a retracted state and an engaged state. Each morphable bag support comprises fingers extending away from the support face and separated by gaps, and pressure chambers within the fingers connected by an internal channel.

(9) The features and functions can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments in which further details can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features believed characteristic of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, however, as well as a preferred mode of use, further objectives and features thereof, will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings, wherein:

(2) FIG. 1 is an illustration of an aircraft in accordance with an illustrative embodiment;

(3) FIG. 2 is an illustration of a block diagram of a manufacturing environment in accordance with an illustrative embodiment;

(4) FIG. 3 is an illustration of a composite compaction system with a number of morphable bag supports in a retracted state in accordance with an illustrative embodiment;

(5) FIG. 4 is an illustration of a composite compaction system with a number of morphable bag supports in an engaged state in accordance with an illustrative embodiment;

(6) FIG. 5 is an illustration of a composite compaction system with a number of morphable bag supports in a retracted state against an object in accordance with an illustrative embodiment;

(7) FIG. 6 is an illustration of a composite compaction system with a number of morphable bag supports in an engaged state against an object in accordance with an illustrative embodiment;

(8) FIG. 7 is an illustration of a composite compaction system with a number of morphable bag supports in a retracted state in accordance with an illustrative embodiment;

(9) FIG. 8 is an illustration of a top view of a composite compaction system with a number of morphable bag supports in accordance with an illustrative embodiment;

(10) FIG. 9 is an illustration of a top view of a composite compaction system with a number of morphable bag supports in accordance with an illustrative embodiment;

(11) FIG. 10 is an illustration of a top view of a composite compaction system with a number of

morphable bag supports in accordance with an illustrative embodiment;

(12) FIG. **11** is a flowchart of a method of supporting a composite compaction system in accordance with an illustrative embodiment;

(13) FIG. **12** is an illustration of an aircraft manufacturing and service method in a form of a block diagram in accordance with an illustrative embodiment; and

(14) FIG. **13** is an illustration of an aircraft in a form of a block diagram in which an illustrative embodiment may be implemented.

DETAILED DESCRIPTION

(15) The illustrative examples recognize and take into account one or more different considerations. The illustrative examples recognize and take into account that existing automated solutions for composite compaction involve the use of linear actuators to support and manipulate compaction films from a single contact point. The illustrative examples recognize and take into account that the single contact point can provide an undesirably low amount of support of the film. The illustrative examples recognize and take into account that the rigidity of the mechanism can prevent the compaction film from adequate seating on contoured surfaces.

(16) The illustrative examples recognize and take into account that current compaction bag supports present some issues. The illustrative examples recognize and take into account that current compaction bag supports present insufficient support of compaction bag due to attachment points located only at corners. The illustrative examples recognize and take into account that current compaction bag supports provide a lack of conformity and adequate sealing of compaction bag to contoured surface due to current rigid supports. The illustrative examples also recognize and take into account that undesirably restraining the bag using the current compaction bag supports could undesirably impact the composite structure underneath.

(17) Turning now to FIG. **1**, an illustration of an aircraft is depicted in accordance with an illustrative embodiment. Aircraft **100** has wing **102** and wing **104** attached to body **106**. Aircraft **100** includes engine **108** attached to wing **102** and engine **110** attached to wing **104**.

(18) Body **106** has tail section **112**. Horizontal stabilizer **114**, horizontal stabilizer **116**, and vertical stabilizer **118** are attached to tail section **112** of body **106**.

(19) Aircraft **100** is an example of an aircraft having composite components that can be manufactured using composite compaction system of the illustrative examples.

(20) Turning now to FIG. **2**, an illustration of a block diagram of a manufacturing environment is depicted in accordance with an illustrative embodiment. Composite compaction system **200** in manufacturing environment **202** can be used to compact material **203** against support structure **206**. In some illustrative examples, material **203** takes the form of composite material **204**. In some illustrative examples, support structure **206** takes the form of a mandrel or other type of tool. In some illustrative examples, support structure **206** takes the form of a substrate, such as a portion of a part.

(21) Composite compaction system **200** comprises vacuum bag **208** having contact face **210** and support face **212** opposite contact face **210**. Contact face **210** is configured to contact material **203**. In some illustrative examples, contact face **210** is configured to contact composite material **204**. Composite compaction system **200** also comprises number of morphable bag supports **214** connected to support face **212** and configured to morph between retracted state **218** and engaged state **220**. In retracted state **218**, number of morphable bag supports **214** can lift corners **222** of vacuum bag **208**. In retracted state **218**, number of morphable bag supports **214** hold a portion of vacuum bag **208** above support structure **206**. In some illustrative examples, number of morphable bag supports **214** is formed of flexible polymeric material **216**.

(22) In some illustrative examples, each of number of morphable bag supports **214** has a same design. In some illustrative examples, at least one of number of morphable bag supports **214** has a different design than another morphable bag support of number of morphable bag supports **214**.

(23) Each morphable bag support of number of morphable bag supports **214** comprises fingers

separated by gaps, the fingers extending outwardly away from the vacuum bag. For example, first morphable bag support **224** comprises fingers **226** separated by gaps **228**, fingers **226** extending outwardly away from vacuum bag **208**. Second morphable bag support **230** comprises fingers **232** separated by gaps **234**, fingers **232** extending outwardly away from vacuum bag **208**.

(24) Each morphable bag support of number of morphable bag supports **214** further comprises pressure chambers within the fingers connected by an internal channel. For example, first morphable bag support **224** comprises pressure chambers **236** within fingers **226** connected by internal channel **238**. Second morphable bag support **230** comprises pressure chambers **240** within fingers **232** connected by internal channel **242**.

(25) Number of morphable bag supports **214** are actuated between engaged state **220** and retracted state **218** by modifying pressure within the pressure chambers of number of morphable bag supports **214**. For example, first pressure source **244** is connected to pressure input **246** of first morphable bag support **224**. First pressure source **244** can provide either vacuum **248** or positive pressure **250**.

(26) When first pressure source **244** provides vacuum **248** to pressure chambers **236** of first morphable bag support **224**, first morphable bag support **224** is in retracted state **218**. When first pressure source **244** provides positive pressure **250** to pressure chambers **236** of first morphable bag support **224**, first morphable bag support **224** is in engaged state **220**.

(27) Second pressure source **252** is connected to pressure input **254** of second morphable bag support **230**. second pressure source **252** can provide either vacuum **256** or positive pressure **258**.

(28) When second pressure source **252** provides vacuum **256** to pressure chambers **240** of second morphable bag support **230**, second morphable bag support **230** is in retracted state **218**. When first pressure source **244** provides positive pressure **258** to pressure chambers **240** of second morphable bag support **230**, second morphable bag support **230** is in engaged state **220**.

(29) In some illustrative examples, each of number of morphable bag supports **214** is independently controlled. For example, first morphable bag support **224** is controlled independently of second morphable bag support **230**.

(30) A rate of contraction of first morphable bag support **224** is controlled by at least one of pressure supplied by first pressure source **244**, width **260** of gaps **228**, and height **262** of fingers **226**. Height **262** limits contraction of first morphable bag support **224**. Increasing height **262** reduces the contraction of first morphable bag support **224**. Increasing width **260** of gaps **228** allows for more movement of first morphable bag support **224**. Moment of inertia of first morphable bag support **224** is improved with increased height **262**. Flexibility of first morphable bag support **224** is increased with decreased height **262**.

(31) In some illustrative examples, height **262** is consistent for fingers **226**. In some illustrative examples, height **262** varies for fingers **226**. In some illustrative examples, pressure chambers **236** are the same size and shape. In some illustrative examples, at least one pressure chamber of pressure chambers **236** can have a different size and shape.

(32) A rate of contraction of second morphable bag support **230** is controlled by at least one of pressure supplied by second pressure source **252**, width **266** of gaps **234**, and height **268** of fingers **232**. Height **268** limits contraction of second morphable bag support **230**. Increasing height **268** reduces the contraction of second morphable bag support **230**. Increasing width **266** of gaps **234** allows for more movement of second morphable bag support **230**. Moment of inertia of second morphable bag support **230** is improved with increased height **268**. Flexibility of second morphable bag support **230** is increased with decreased height **268**.

(33) In some illustrative examples, height **268** is consistent for fingers **232**. In some illustrative examples, height **268** varies for fingers **232**. In some illustrative examples, pressure chambers **240** are the same size and shape. In some illustrative examples, at least one pressure chamber of pressure chambers **240** can have a different size and shape.

(34) Fingers **226** extend outwardly from base **270**. Base **270** of first morphable bag support **224** is

connected to support face **212** of vacuum bag **208**. In some illustrative examples, each of number of morphable bag supports **214** is connected to support face **212** on periphery **272** of vacuum bag **208**. In some illustrative examples, first morphable bag support **224** is connected to first side **274** of support face **212**.

(35) Fingers **232** extend outwardly from base **276**. Base **276** of second morphable bag support **230** is connected to support face **212** of vacuum bag **208**. In some illustrative examples, second morphable bag support **230** is connected to second side **278** of support face **212**.

(36) In this illustrative example, number of morphable bag supports **214** comprises first morphable bag support **224** connected to first side **274** of support face **212** and second morphable bag support **230** connected to second side **278** of support face **212** such that four corners **222** of vacuum bag **208** are movable by actuating first morphable bag support **224** and second morphable bag support **230**. In some illustrative examples, number of morphable bag supports **214** further comprises other morphable bag supports not depicted in FIG. **2**. In some illustrative examples, number of morphable bag supports **214** further comprises a third morphable bag support (not depicted) connected to third side **284** of support face **212** connecting first side **274** and second side **278** and a fourth morphable bag support (not depicted) connected to fourth side **286** of support face **212** connecting first side **274** and second side **278**.

(37) Number of morphable bag supports **214** are utilized for supporting vacuum bag **208** of composite compaction system **200** to place composite compaction system **200** on support structure **206**. Number of morphable bag supports **214** are utilized to conform vacuum bag **208** to support structure **206** prior to compacting material **203**.

(38) Composite compaction system **200** further comprises vacuum interface structure **280** configured to control vacuum **281** supplied to vacuum bag **208** for compaction **283** of material **203**. Compaction pressure source **282** provides pressure control to vacuum interface structure **280** independently of pressure control on number of morphable bag supports **214**. In some illustrative examples, vacuum interface structure **280** is centered relative to support face **212** of vacuum bag **208**.

(39) Composite compaction system **200** further comprises rigid support **288** connected to vacuum interface structure **280** and number of morphable bag supports **214**. Although one rigid support, rigid support **288** is depicted, any desirable quantity of rigid supports is present to connect number of morphable bag supports **214** to vacuum interface structure **280**.

(40) As depicted, composite compaction system **200** comprises number of pressure sources **290** having a quantity equivalent to a quantity of morphable bag supports in number of morphable bag supports **214** such that a single morphable bag support of number of morphable bag supports **214** is connected to each pressure source of number of pressure sources **290**. In other non-depicted illustrative examples, number of pressure sources **290** can have a quantity less than the quantity of morphable bag supports in number of morphable bag supports **214**. In these illustrative examples, independently controlled valves can be used to allow more than one morphable bag support to be connected to a single pressure source and also individually controlled.

(41) The illustration of manufacturing environment **202** in FIG. **2** is not meant to imply physical or architectural limitations to the manner in which an illustrative embodiment may be implemented. Other components in addition to or in place of the ones illustrated may be used. Some components may be unnecessary. Also, the blocks are presented to illustrate some functional components. One or more of these blocks may be combined, divided, or combined and divided into different blocks when implemented in an illustrative embodiment.

(42) For example, more than two morphable bag supports can be present in number of morphable bag supports **214**. As another example, more than one rigid support can be present.

(43) Turning now to FIG. **3**, an illustration of a composite compaction system with a number of morphable bag supports in a retracted state is depicted in accordance with an illustrative embodiment. Composite compaction system **300** is a physical implementation of composite

compaction system **200** of FIG. 2. Composite compaction system **300** can be used to form a portion of aircraft **100** of FIG. 1.

(44) Composite compaction system **300** comprises vacuum bag **302** and number of morphable bag supports **308**. Vacuum bag **302** has contact face **304** and support face **306** opposite contact face **304**. In some illustrative examples, contact face **304** is configured to contact a composite material (not depicted). Number of morphable bag supports **308** is formed of a flexible polymeric material connected to support face **306** and configured to morph between retracted state **310** and an engaged state (not depicted).

(45) In this illustrative example, number of morphable bag supports **308** comprises first morphable bag support **312** connected to first side **314** of support face **306** and second morphable bag support **316** connected to second side **318** of support face **306** such that four corners of vacuum bag **302** are movable by actuating first morphable bag support **312** and second morphable bag support **316**. Each morphable bag support of number of morphable bag supports **308** comprises fingers separated by gaps, the fingers extending outwardly away from vacuum bag **302**.

(46) As depicted, first morphable bag support **312** comprises fingers **322** separated by gaps **324**. Pressure chambers (not depicted) within fingers **322** are connected by an internal channel (not depicted). Pressure supplied to pressure ports **326** control pressure supplied to the pressure chambers (not depicted) within fingers **322**. Application of pressure to the pressure chambers controls the position of first morphable bag support **312**. Application of positive pressure to the pressure chambers via pressure ports **326** changes first morphable bag support **312** between retracted state **310** and an engaged state (not depicted).

(47) As depicted, second morphable bag support **316** comprises fingers **328** separated by gaps **330**. Pressure chambers (not depicted) within fingers **328** are connected by an internal channel (not depicted). Pressure supplied to pressure ports **332** control pressure supplied to the pressure chambers (not depicted) within fingers **328**. Application of pressure to the pressure chambers controls the position of second morphable bag support **316**. Application of positive pressure to the pressure chambers via pressure ports **332** changes second morphable bag support **316** between retracted state **310** and an engaged state (not depicted).

(48) Composite compaction system **300** further comprises vacuum interface structure **334**. Vacuum interface structure **334** is configured to control vacuum supplied to vacuum bag **302** for compaction of a composite material. Vacuum is supplied to vacuum interface structure **334** independently of pressure supplied to number of morphable bag supports **308**.

(49) In this illustrative example, rigid support **336** is connected to vacuum interface structure **334** and number of morphable bag supports **308**. Rigid support **336** provides structure to composite compaction system **300**.

(50) Turning now to FIG. 4, an illustration of a composite compaction system with a number of morphable bag supports in an engaged state is depicted in accordance with an illustrative embodiment. View **400** is a view of composite compaction system **300** of FIG. 3 with number of morphable bag supports **308** in engaged state **402**. In engaged state **402**, number of morphable bag supports **308** are shaped such that vacuum bag **302** is configured to contact a surface of an object (not depicted).

(51) Turning now to FIG. 5, an illustration of a composite compaction system with a number of morphable bag supports in a retracted state against an object is depicted in accordance with an illustrative embodiment. View **500** is an isometric side view of composite compaction system **300** in contact with object **502** while number of morphable bag supports **308** is in retracted state **310**. In view **500**, only a portion of vacuum bag **302** is in contact with surface **504** of object **502**.

(52) In view **500**, corner **506** and corner **508** of vacuum bag **302** are lifted off of object **502**. In view **500**, retracted state **310** reduces wrinkling of vacuum bag **302** when positioning composite compaction system **300** against object **502**.

(53) In view **500**, a vacuum is applied to number of morphable bag supports **308** in order to lift

corner **506** and corner **508** of vacuum bag **302**. Composite compaction system **300** is transferred with number of morphable bag supports **308** in retracted state **310** and placed over object **502**.

(54) In view **500**, surface **504** of object **502** is a contoured surface. It would be undesirably difficult to apply a rigidly supported structure to contour **510** of surface **504**. Number of morphable bag supports **308** allows for composite compaction system **300** to engage surface **504** of object **502**, as shown in FIG. **6** below.

(55) Turning now to FIG. **6**, an illustration of a composite compaction system with a number of morphable bag supports in an engaged state against an object is depicted in accordance with an illustrative embodiment. In view **600**, composite compaction system **300** is positioned with vacuum bag **302** against object **502**. In view **600**, number of morphable bag supports **308** is in engaged state **402**. In view **600** corner **506** and corner **508** of vacuum bag **302** are in contact with surface **504** of object **502**. Vacuum bag **302** is positioned against object **502** with reduced wrinkling due to number of morphable bag supports **308**.

(56) Between view **500** and view **600**, the vacuum has been released from number of morphable bag supports **308**. Afterwards, positive pressure is applied to number of morphable bag supports **308** to conform vacuum bag **302** to surface **504** of object **502**. In some illustrative examples, the positive pressure takes the form of compressed air. Number of morphable bag supports **308** provide adequate support and seating when conforming to the shape of surface **504**.

(57) Turning now to FIG. **7**, an illustration of a composite compaction system with a number of morphable bag supports in a retracted state is depicted in accordance with an illustrative embodiment. Composite compaction system **700** is a physical implementation of composite compaction system **200** of FIG. **2**. Composite compaction system **700** can be used to form a portion of aircraft **100** of FIG. **1**.

(58) Composite compaction system **700** comprises vacuum bag **702** and number of morphable bag supports **708**. Vacuum bag **702** has contact face **704** and support face **706** opposite contact face **704**. In some illustrative examples, contact face **704** is configured to contact a composite material (not depicted). Number of morphable bag supports **708** is formed of a flexible polymeric material connected to support face **706** and configured to morph between retracted state **710** and an engaged state (not depicted).

(59) In this illustrative example, number of morphable bag supports **708** comprises first morphable bag support **712** connected to first side **714** of support face **706**, second morphable bag support **716** connected to second side **718** of support face **706**, third morphable bag support **711** connected to third side **707** of support face **706**, and fourth morphable bag support **719** connected to fourth side **709** of support face **706** such that four corners of vacuum bag **702** are movable by actuating number of morphable bag supports **708**. Each morphable bag support of number of morphable bag supports **708** comprises fingers separated by gaps, the fingers extending outwardly away from vacuum bag **702**.

(60) As depicted, first morphable bag support **712** comprises fingers **722** separated by gaps **724**. Pressure chambers (not depicted) within fingers **722** are connected by an internal channel (not depicted). Pressure supplied to pressure ports **726** control pressure supplied to the pressure chambers (not depicted) within fingers **722**. Application of pressure to the pressure chambers controls the position of first morphable bag support **712**. Application of positive pressure to the pressure chambers via pressure ports **726** changes first morphable bag support **712** between retracted state **710** and an engaged state (not depicted).

(61) As depicted, second morphable bag support **716** comprises fingers **728** separated by gaps **730**. Pressure chambers (not depicted) within fingers **728** are connected by an internal channel (not depicted). Pressure supplied to pressure ports **732** control pressure supplied to the pressure chambers (not depicted) within fingers **728**. Application of pressure to the pressure chambers controls the position of second morphable bag support **716**. Application of positive pressure to the pressure chambers via pressure ports **732** changes second morphable bag support **716** between

retracted state **710** and an engaged state (not depicted).

(62) As depicted, third morphable bag support **711** comprises fingers **713** separated by gaps **715**. Pressure chambers (not depicted) within fingers **713** are connected by an internal channel (not depicted). Pressure supplied to pressure ports **717** control pressure supplied to the pressure chambers (not depicted) within fingers **713**. Application of pressure to the pressure chambers controls the position of third morphable bag support **711**. Application of positive pressure to the pressure chambers via pressure ports **717** changes third morphable bag support **711** between retracted state **710** and an engaged state (not depicted).

(63) As depicted, fourth morphable bag support **719** comprises fingers **721** separated by gaps **723**. Pressure chambers (not depicted) within fingers **721** are connected by an internal channel (not depicted). Pressure supplied to pressure ports **725** control pressure supplied to the pressure chambers (not depicted) within fingers **721**. Application of pressure to the pressure chambers controls the position of fourth morphable bag support **719**. Application of positive pressure to the pressure chambers via pressure ports **725** changes fourth morphable bag support **719** between retracted state **710** and an engaged state (not depicted).

(64) Composite compaction system **700** further comprises vacuum interface structure **734**. Vacuum interface structure **734** is configured to control vacuum supplied to vacuum bag **702** for compaction of a composite material. Vacuum is supplied to vacuum interface structure **734** independently of pressure supplied to number of morphable bag supports **708**.

(65) In this illustrative example, rigid support **736** is connected to vacuum interface structure **734**, first morphable bag support **712** and second morphable bag support **716**. Rigid support **736** provides structure to composite compaction system **700**. In this illustrative example, rigid support **736** is connected to vacuum interface structure **734**, third morphable bag support **711**, and fourth morphable bag support **719**. Rigid support **736** provides structure to composite compaction system **700**.

(66) Turning now to FIG. **8**, an illustration of a top view of a composite compaction system with a number of morphable bag supports is depicted in accordance with an illustrative embodiment. Composite compaction system **800** is a physical implementation of composite compaction system **200** of FIG. **2**. Composite compaction system **800** can be used to form a portion of aircraft **100** of FIG. **1**.

(67) Composite compaction system **800** comprises vacuum bag **802** and number of morphable bag supports **804**. Vacuum bag **802** has contact face (not depicted) and support face **806** opposite the contact face. In some illustrative examples, the contact face is configured to contact a composite material (not depicted). Number of morphable bag supports **804** is formed of a flexible polymeric material connected to support face **806** and configured to morph between a retracted state and an engaged state.

(68) Each morphable bag support of number of morphable bag supports **804** comprises fingers separated by gaps, the fingers extending outwardly away from vacuum bag **802**. Pressure chambers of each morphable bag support of number of morphable bag supports **804** are connected by a respective internal channel (not depicted). Pressure supplied to pressure ports of number of morphable bag supports **804** controls pressure supplied to the pressure chambers (not depicted) within the respective fingers. Application of pressure to the respective pressure chambers controls the position of a respective morphable bag support of number of morphable bag supports **804**. Application of positive pressure to respective pressure chambers via pressure ports changes a respective morphable bag support between a retracted state and an engaged state.

(69) Composite compaction system **800** further comprises vacuum interface structure **808**. Vacuum interface structure **808** is configured to control vacuum supplied to vacuum bag **802** for compaction of a composite material. Vacuum is supplied to vacuum interface structure **808** independently of pressure supplied to number of morphable bag supports **804**.

(70) In this illustrative example, vacuum bag **802** is circular **810**. In this illustrative example,

number of morphable bag supports **804** are connected to vacuum interface structure **808**. In this illustrative example, number of morphable bag supports **804** are laid out radially **812** from vacuum interface structure **808**.

(71) In this illustrative example, although number of morphable bag supports **804** comprises eight morphable bag supports, any desirable quantity of morphable bag supports can be present. In some illustrative examples, number of morphable bag supports **804** can have fewer than eight morphable bag supports. In some illustrative examples, number of morphable bag supports **804** can have more than eight morphable bag supports.

(72) In this illustrative example, number of morphable bag supports **804** extends from vacuum interface structure **808** to edge **814** of vacuum bag **802**. In some other non-depicted illustrative examples, number of morphable bag supports **804** terminates prior to edge **814** of vacuum bag **802**. In this illustrative example, each of number of morphable bag supports **804** has a same length. In other non-depicted illustrative examples, at least one morphable bag support of number of morphable bag supports **804** has a different length than other morphable bag supports in number of morphable bag supports **804**.

(73) Turning now to FIG. **9**, an illustration of a top view of a composite compaction system with a number of morphable bag supports is depicted in accordance with an illustrative embodiment. Composite compaction system **900** is a physical implementation of composite compaction system **200** of FIG. **2**. Composite compaction system **900** can be used to form a portion of aircraft **100** of FIG. **1**.

(74) Composite compaction system **900** comprises vacuum bag **902** and number of morphable bag supports **904**. Vacuum bag **902** has contact face (not depicted) and support face **906** opposite the contact face. In some illustrative examples, the contact face is configured to contact a composite material (not depicted). Number of morphable bag supports **904** is formed of a flexible polymeric material connected to support face **906** and configured to morph between a retracted state and an engaged state.

(75) Each morphable bag support of number of morphable bag supports **904** comprises fingers separated by gaps, the fingers extending outwardly away from vacuum bag **902**. Pressure chambers of each morphable bag support of number of morphable bag supports **904** are connected by a respective internal channel (not depicted). Pressure supplied to pressure ports of number of morphable bag supports **904** controls pressure supplied to the pressure chambers (not depicted) within the respective fingers. Application of pressure to the respective pressure chambers controls the position of a respective morphable bag support of number of morphable bag supports **904**. Application of positive pressure to respective pressure chambers via pressure ports changes a respective morphable bag support between a retracted state and an engaged state.

(76) Composite compaction system **900** further comprises vacuum interface structure **908**. Vacuum interface structure **908** is configured to control vacuum supplied to vacuum bag **902** for compaction of a composite material. Vacuum is supplied to vacuum interface structure **908** independently of pressure supplied to number of morphable bag supports **904**.

(77) In this illustrative example, vacuum bag **902** is octagonal **910**. In this illustrative example, number of morphable bag supports **904** are connected to vacuum interface structure **908**. In this illustrative example, number of morphable bag supports **904** are laid out radially **912** from vacuum interface structure **908**.

(78) In this illustrative example, although number of morphable bag supports **904** comprises eight morphable bag supports, any desirable quantity of morphable bag supports can be present. In some illustrative examples, number of morphable bag supports **904** can have fewer than eight morphable bag supports. In some illustrative examples, number of morphable bag supports **904** can have more than eight morphable bag supports.

(79) In this illustrative example, number of morphable bag supports **904** extends from vacuum interface structure **908** to edge **914** of vacuum bag **902**. In this illustrative example, number of

morphable bag supports **904** extends from vacuum interface structure **908** to number of corners **916** of vacuum bag **902**. In some other non-depicted illustrative examples, number of morphable bag supports **904** terminates prior to edge **914** of vacuum bag **902**. In this illustrative example, each of number of morphable bag supports **904** has a same length. In other non-depicted illustrative examples, at least one morphable bag support of number of morphable bag supports **904** has a different length than other morphable bag supports in number of morphable bag supports **904**.

(80) Turning now to FIG. **10**, an illustration of a top view of a composite compaction system with a number of morphable bag supports is depicted in accordance with an illustrative embodiment. Composite compaction system **1000** is a physical implementation of composite compaction system **200** of FIG. **2**. Composite compaction system **1000** can be used to form a portion of aircraft **100** of FIG. **1**.

(81) Composite compaction system **1000** comprises vacuum bag **1002** and number of morphable bag supports **1004**. Vacuum bag **1002** has contact face (not depicted) configured to contact a material (not depicted) and support face **1006** opposite contact face. Number of morphable bag supports **1004** is formed of a flexible polymeric material connected to support face **1006** and configured to morph between a retracted state and an engaged state.

(82) Each morphable bag support of number of morphable bag supports **1004** comprises fingers separated by gaps, the fingers extending outwardly away from vacuum bag **1002**. Pressure chambers of each morphable bag support of number of morphable bag supports **1004** are connected by a respective internal channel (not depicted). Pressure supplied to pressure ports of number of morphable bag supports **1004** controls pressure supplied to the pressure chambers (not depicted) within the respective fingers. Application of pressure to the respective pressure chambers controls the position of a respective morphable bag support of number of morphable bag supports **1004**. Application of positive pressure to respective pressure chambers via pressure ports changes a respective morphable bag support between a retracted state and an engaged state.

(83) Composite compaction system **1000** further comprises vacuum interface structure **1008**. Vacuum interface structure **1008** is configured to control vacuum supplied to vacuum bag **1002** for compaction of a composite material. Vacuum is supplied to vacuum interface structure **1008** independently of pressure supplied to number of morphable bag supports **1004**.

(84) In this illustrative example, vacuum bag **1002** is hexagonal **1010**. In this illustrative example, number of morphable bag supports **1004** is parallel to number of sides **1012** of edge **1014**.

(85) In this illustrative example, although number of morphable bag supports **1004** comprises six morphable bag supports, any desirable quantity of morphable bag supports can be present. In some illustrative examples, number of morphable bag supports **1004** can have fewer than six morphable bag supports. In some illustrative examples, three morphable bag supports are present in number of morphable bag supports **1004**, parallel to alternating sides of number of sides **1012**. In some illustrative examples, number of morphable bag supports **1004** can have more than six morphable bag supports.

(86) In this illustrative example, number of rigid supports **1016** extend from vacuum interface structure **1008** to number of morphable bag supports **1004**. In this illustrative example, number of rigid supports **1016** extend radially **1018** from vacuum interface structure **1008** to number of morphable bag supports **1004**.

(87) Turning now to FIG. **11**, a flowchart of a method of supporting a composite compaction system is depicted in accordance with an illustrative embodiment. Method **1100** can be used to process a composite portion of aircraft **100** of FIG. **1**. Method **1100** can be performed using composite compaction system **200** of FIG. **2**. Method **1100** can be performed using composite compaction system **300** of FIGS. **3-6**. Method **1100** can be performed using composite compaction system **700** of FIG. **7**.

(88) Method **1100** holds a number of morphable bag supports of the composite compaction system in a retracted state (operation **1102**). Method **1100** places a contact face of a vacuum bag of the

composite compaction system in contact with a material on a support structure (operation **1104**). Method **1100** places the number of morphable bag supports in an engaged state to conform to the support structure (operation **1106**). Afterwards, method **1100** terminates.

(89) In some illustrative examples, holding the number of morphable bag supports in the retracted state comprises applying vacuum to pressure chambers within the number of morphable bag supports (operation **1108**). In some illustrative examples, applying the vacuum to the pressure chambers within the number of morphable bag supports comprises applying a respective vacuum to each morphable bag support of the number of morphable bag supports independently of each other morphable bag support (operation **1110**).

(90) In some illustrative examples, placing the number of morphable bag supports in the engaged state comprises applying a positive pressure to pressure chambers within the number of morphable bag supports (operation **1112**). In some illustrative examples, applying the positive pressure to the pressure chambers within the number of morphable bag supports comprises applying positive pressure to each morphable bag support of the number of morphable bag supports independently of each other morphable bag support (operation **1114**).

(91) In some illustrative examples, method **1100** further comprises performing compaction of the material on the support structure using the composite compaction system (operation **1116**). In some illustrative examples, performing compaction comprises applying a vacuum beneath the vacuum bag (operation **1118**).

(92) In some illustrative examples, method **1100** places the number of morphable bag supports in the retracted state after performing compaction of the material (operation **1120**). After placing the number of morphable bag supports in the retracted state, the composite compaction system can be lifted away from the composite material on the support structure.

(93) As used herein, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items may be used and only one of each item in the list may be needed. For example, “at least one of item A, item B, or item C” may include, without limitation, item A, item A and item B, or item B. This example also may include item A, item B, and item C or item B and item C. Of course, any combinations of these items may be present. In other examples, “at least one of” may be, for example, without limitation, two of item A; one of item B; and ten of item C; four of item B and seven of item C; or other suitable combinations. The item may be a particular object, thing, or a category. In other words, at least one of means any combination items and number of items may be used from the list but not all of the items in the list are required.

(94) As used herein, “a number of,” when used with reference to items means one or more items.

(95) The flowcharts and block diagrams in the different depicted embodiments illustrate the architecture, functionality, and operation of some possible implementations of apparatuses and methods in an illustrative embodiment. In this regard, each block in the flowcharts or block diagrams may represent at least one of a module, a segment, a function, or a portion of an operation or step.

(96) In some alternative implementations of an illustrative embodiment, the function or functions noted in the blocks may occur out of the order noted in the figures. For example, in some cases, two blocks shown in succession may be executed substantially concurrently, or the blocks may sometimes be performed in the reverse order, depending upon the functionality involved. Also, other blocks may be added in addition to the illustrated blocks in a flowchart or block diagram. Some blocks may be optional. For example, operation **1108** through operation **1120** may be optional.

(97) Illustrative embodiments of the present disclosure may be described in the context of aircraft manufacturing and service method **1200** as shown in FIG. **12** and aircraft **1300** as shown in FIG. **13**. Turning first to FIG. **12**, an illustration of an aircraft manufacturing and service method in a form of a block diagram is depicted in accordance with an illustrative embodiment. During pre-

production, aircraft manufacturing and service method **1200** may include specification and design **1202** of aircraft **1300** in FIG. **13** and material procurement **1204**.

(98) During production, component and subassembly manufacturing **1206** and system integration **1208** of aircraft **1300** takes place. Thereafter, aircraft **1300** may go through certification and delivery **1210** in order to be placed in service **1212**. While in service **1212** by a customer, aircraft **1300** is scheduled for routine maintenance and service **1214**, which may include modification, reconfiguration, refurbishment, or other maintenance and service.

(99) Each of the processes of aircraft manufacturing and service method **1200** may be performed or carried out by a system integrator, a third party, and/or an operator. In these examples, the operator may be a customer. For the purposes of this description, a system integrator may include, without limitation, any number of aircraft manufacturers and major-system subcontractors; a third party may include, without limitation, any number of vendors, subcontractors, and suppliers; and an operator may be an airline, a leasing company, a military entity, a service organization, and so on.

(100) With reference now to FIG. **13**, an illustration of an aircraft in a form of a block diagram is depicted in which an illustrative embodiment may be implemented. In this example, aircraft **1300** is produced by aircraft manufacturing and service method **1200** of FIG. **12** and may include airframe **1302** with plurality of systems **1304** and interior **1306**. Examples of systems **1304** include one or more of propulsion system **1308**, electrical system **1310**, hydraulic system **1312**, and environmental system **1314**. Any number of other systems may be included.

(101) Apparatuses and methods embodied herein may be employed during at least one of the stages of aircraft manufacturing and service method **1200**. One or more illustrative embodiments may be manufactured or used during at least one of component and subassembly manufacturing **1206**, system integration **1208**, in service **1212**, or maintenance and service **1214** of FIG. **12**.

(102) A portion of airframe **1302** of aircraft **1300** can be formed using method **1100**. A portion of airframe **1302** of aircraft **1300** can be formed using composite compaction system **200** of FIG. **2**. Method **1100** can be performed during component and subassembly manufacturing **1206**. A composite structure formed using composite compaction system **200** can be present and utilized during in service **1212**. Composite compaction system **200** can be used during maintenance and service **1214** to form a replacement part. Method **1100** can be used during maintenance and service **1214** to form a replacement part.

(103) The illustrative examples provide an apparatus that enables automated and controlled manipulation, support, and seating of compaction films used on composite fabrication processes. The illustrative examples utilize morphable bag supports. The morphable bag supports are attached to the compaction film along the periphery, providing adequate support and seating when conforming to the shape of the surface. In some illustrative examples, morphable supports are attached to the compaction film along the entire periphery.

(104) The illustrative examples utilize a long morphable support made of elastic material containing a series of consecutive air chambers connected with a small air channel. Applied air pressure allows the morphable support to be morphed in one direction with positive pressure and morphed in opposite direction with negative pressure.

(105) The illustrative examples provide continuous support and manipulation of the compaction film while providing some compliance during seating. The technical features of the illustrative examples include compliance with the contoured surface, continuous support of the compaction film, and simplicity of the actuated mechanism.

(106) These long elastic parts, referred to as morphable bag supports, are attached to the compaction film along the periphery, providing adequate support and seating when conforming to the shape of the surface.

(107) The illustrative examples incorporate compliance with the contoured surface of the composite material, continuous support due to attachment to periphery of compaction bag, and simplicity of actuated mechanism. The elastic material of the morphable bag supports reduce the

likeliness to crush or manipulate contour of composite material.

(108) The illustrative examples apply vacuum to the number of morphable bag supports in order to lift corners of bag. The illustrative examples transfer and place a compaction device on a composite material. The illustrative examples release vacuum applied to the number of morphable bag supports. The illustrative examples apply positive pressure (can take the form of compressed air) to the number of morphable bag supports to conform the compaction bag to the composite material contoured surface. The illustrative examples apply initial vacuum to the compaction bag.

(109) The illustrative examples provide continuous morphable support for an automated compaction bag. Through the use of positive and negative air pressure, the illustrative examples can manipulate orientation that will assist with retraction and seating of compaction bag.

(110) Benefits of using the illustrative examples include increased quality. The illustrative examples minimize risk of insufficient compaction due to improved seating of bag. The illustrative examples minimize risk of undesirable effects to composite material due to elastic conformity to part.

(111) The description of the different illustrative embodiments has been presented for purposes of illustration and description, and is not intended to be exhaustive or limited to the embodiments in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different illustrative embodiments may provide different features as compared to other illustrative embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principles of the embodiments, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A method of supporting a composite compaction system comprising: holding a number of morphable bag supports of the composite compaction system in a retracted state; placing a contact face of a vacuum bag of the composite compaction system in contact with a material on a support structure; and placing the number of morphable bag supports in an engaged state to conform to the support structure, wherein each of the number of morphable bag supports comprises a plurality of fingers separated by gaps and a plurality of pressure chambers within the plurality of fingers, wherein holding the number of morphable bag supports in the retracted state comprises applying a vacuum to the plurality of pressure chambers within the plurality of fingers, and wherein placing the number of morphable bag supports in the engaged state comprises applying a positive pressure to the plurality of pressure chambers within the plurality of fingers, wherein applying the vacuum to the plurality of pressure chambers within each of the number of morphable bag supports comprises applying a respective vacuum to each morphable bag support of the number of morphable bag supports independently of each other morphable bag support, and wherein applying the positive pressure to the plurality of pressure chambers within each of the number of morphable bag supports comprises applying positive pressure to each morphable bag support of the number of morphable bag supports independently of each other morphable bag support.
2. The method of claim 1 further comprising: performing compaction of the material on the support structure using the composite compaction system.
3. The method of claim 2, wherein performing compaction comprises applying a vacuum beneath the vacuum bag.
4. The method of claim 2 further comprising: placing the number of morphable bag supports in the retracted state after performing compaction of the material.
5. The method of claim 1 further comprising: seating the vacuum bag on a contoured surface of the support structure after placing the number of morphable bag supports in the engaged state to conform to the support structure.

6. A method of supporting a composite compaction system comprising: holding a plurality of morphable bag supports of the composite compaction system in a retracted state; placing a contact face of a vacuum bag of the composite compaction system in contact with a material on a support structure; and placing the plurality of morphable bag supports in an engaged state to conform to the support structure, wherein each of the plurality of morphable bag supports comprises a plurality of fingers separated by gaps and a plurality of pressure chambers within the plurality of fingers, wherein holding the plurality of morphable bag supports in the retracted state comprises applying a vacuum to the plurality of pressure chambers within the plurality of fingers, and wherein placing the plurality of morphable bag supports in the engaged state comprises applying a positive pressure to the plurality of pressure chambers within the plurality of finger, wherein applying the vacuum to the plurality of pressure chambers within each of the plurality of morphable bag supports comprises applying a respective vacuum to each morphable bag support of the plurality of morphable bag supports independently of each other morphable bag support, and wherein applying the positive pressure to the plurality of pressure chambers within each of the plurality of morphable bag supports comprises applying positive pressure to each morphable bag support of the plurality of morphable bag supports independently of each other morphable bag support.
7. The method of claim 6 further comprising: performing compaction of the material on the support structure using the composite compaction system.
8. The method of claim 7, wherein performing compaction comprises applying a vacuum beneath the vacuum bag.
9. The method of claim 7 further comprising: placing the plurality of morphable bag supports in the retracted state after performing compaction of the material.
10. The method of claim 6 further comprising: seating the vacuum bag on a contoured surface of the support structure after placing the plurality of morphable bag supports in the engaged state to conform to the support structure.
11. A method of supporting a composite compaction system comprising: holding four morphable bag supports of the composite compaction system in a retracted state; placing a contact face of a vacuum bag of the composite compaction system in contact with a material on a support structure; and placing the four morphable bag supports in an engaged state to conform to the support structure, wherein each of the four morphable bag supports comprises a plurality of fingers separated by gaps and a plurality of pressure chambers within the plurality of fingers, wherein holding the four morphable bag supports in the retracted state comprises applying a vacuum to the plurality of pressure chambers within the plurality of fingers, and wherein placing the four morphable bag supports in the engaged state comprises applying a positive pressure to the plurality of pressure chambers within the plurality of fingers, wherein applying the vacuum to the plurality of pressure chambers within each of the four morphable bag supports comprises applying a respective vacuum to each morphable bag support of the four morphable bag supports independently of each other morphable bag support, and wherein applying the positive pressure to the plurality of pressure chambers within each of the four morphable bag supports comprises applying positive pressure to each morphable bag support of the four morphable bag supports independently of each other morphable bag support.
12. The method of claim 11 further comprising: performing compaction of the material on the support structure using the composite compaction system.
13. The method of claim 12, wherein performing compaction comprises applying a vacuum beneath the vacuum bag.
14. The method of claim 12 further comprising: placing the four morphable bag supports in the retracted state after performing compaction of the material.
15. The method of claim 11 further comprising: seating the vacuum bag on a contoured surface of the support structure after placing the four morphable bag supports in the engaged state to conform to the support structure.

16. The method of claim 5, further comprising: performing compaction of the material on the support structure using the composite compaction system.
 17. The method of claim 16, wherein performing compaction comprises applying a vacuum beneath the vacuum bag.
 18. The method of claim 16, further comprising: placing the number of morphable bag supports in the retracted state after performing compaction of the material.
 19. The method of claim 10, further comprising: performing compaction of the material on the support structure using the composite compaction system.
 20. The method of claim 19, wherein performing compaction comprises applying a vacuum beneath the vacuum bag.
 21. The method of claim 19, further comprising: placing the plurality of morphable bag supports in the retracted state after performing compaction of the material.
 22. The method of claim 15, further comprising: performing compaction of the material on the support structure using the composite compaction system.
 23. The method of claim 22, wherein performing compaction comprises applying a vacuum beneath the vacuum bag.
 24. The method of claim 22, further comprising: placing the four morphable bag supports in the retracted state after performing compaction of the material.
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